Ericsson PE Node BTM-PE-05 Site BTM Layout Aggregation Routing Protocol Flap Failure SOP

Node: BTM-PE-05 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_9945

Description

This SOP outlines the procedures to be followed in the event of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on the Ericsson PE node BTM-PE-05 located at the BTM Layout Aggregation site. The failure domain is a critical aspect of the network's control plane, and a flap can lead to increased latency, packet loss, and potential network instability. The root cause of this failure is often attributed to misconfigurations, hardware issues, or software bugs in the routing protocols or the underlying hardware. In this SOP, we will focus on identifying the root cause of the flap and implementing corrective actions to restore normal network operation. The Ericsson PE node uses the CLI dialect for configuration and troubleshooting, and all commands in this SOP are specified in accordance with the Ericsson CLI syntax.

Pre-requisite Checks

- Verify that the site is accessible and that the network team has been notified.

- Check the redundant link status to ensure that it is functioning correctly.

- Verify that the node is powered on and that the system is stable.

Potential Root Causes

Cause	Probability
Misconfigured routing protocols	High
Hardware issues	Medium
Software bugs	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP router identifier 10.0.0.1, local AS number 65000, Description: This command displays a summary of the BGP routing table, including the number of peers, the number of prefixes, and the number of routes. It helps identify any inconsistencies in the BGP routing table.

Command: show ospf neighbor, Expected Output: Neighbor 10.0.0.2, state FULL, last update 00:00:01, Description: This command displays information about the OSPF neighbors, including their state and the last update time. It helps identify any OSPF adjacency issues.

Command: show isis database, Expected Output: ISIS level 1 database, type Level 1 LSPs, Description: This command displays the ISIS database, including the type of LSPs and the level of the database. It
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```
helps identify any ISIS LSP flooding issues.
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Resolution Steps

Step 1: Step: Verify the routing protocol configuration, Command: show run | begin router bgp, Description: Verify that the routing protocols are configured correctly and that there are no misconfigurations that could be causing the flap.

Step 2: Step: Check the hardware logs, Command: show log hardware, Description: Check the hardware logs for any errors or issues that could be related to the flap.

Step 3: Step: Apply a software patch, Command: install software patch , Description: Apply a software patch to fix any known bugs or issues that could be causing the flap.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP router identifier 10.0.0.1, local AS number 65000, Description: Verify that the BGP routing table is stable and that there are no inconsistencies.

Command: show ospf neighbor, Expected Output: Neighbor 10.0.0.2, state FULL, last update 00:00:01, Description: Verify that the OSPF neighbors are stable and that there are no adjacency issues.

Command: show isis database, Expected Output: ISIS level 1 database, type Level 1 LSPs, Description: Verify that the ISIS database is stable and that there are no LSP flooding issues.
```

Roll-back Procedures

- Step: Revert the configuration changes, Command: copy running-config startup-config, Description: Revert the configuration changes and copy the running configuration to the startup configuration.

- Step: Reset the routing protocols, Command: clear bgp, Description: Reset the BGP routing protocols and clear any stale routes.

- Step: Reboot the node, Command: reboot, Description: Reboot the node to ensure that the changes take effect.